# INTRODUCTION TO DATA SCIENCE

**S. Bosubabu**
Asst. Professor
Department CSE (Data Science)
Mohan Babu University Tirupathi

# LEARNING OBJECTIVES

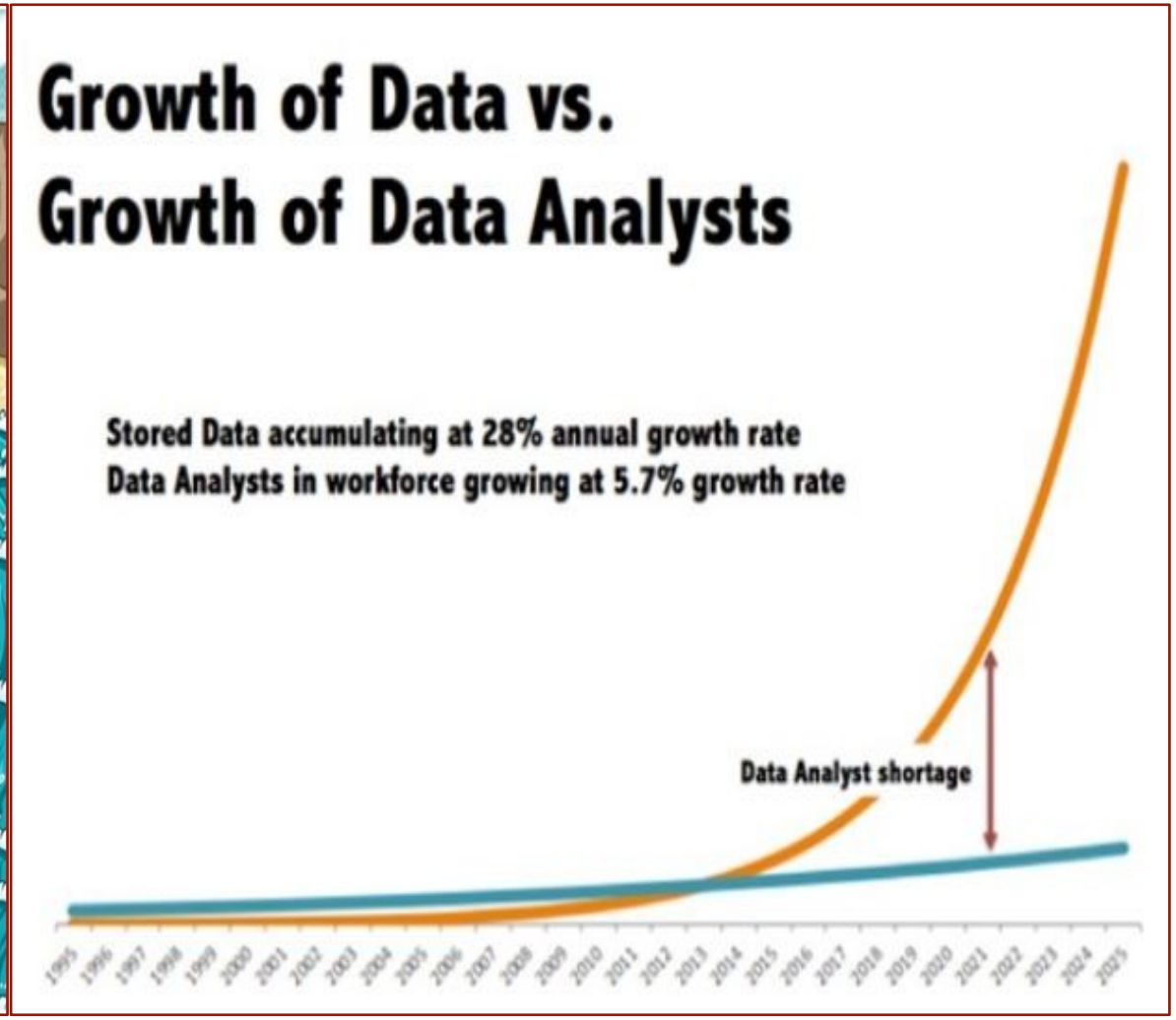
- Apprehend the field of Data Science impact and importance in the society
- Reflect on its applications, importance and advantages

# CONTENTS

- Why should study Data Science?
- How Does Data Science Impact Organizations?
- Application and Competitive Advantage of Data Science in Organization
- Importance of Data Science to Society
- Road to Become a Data Scientist

# WHY WE ARE TALKING ABOUT DATA SCIENCE?






Source: https://bit.ly/31HBHuQ

# WHAT IS DATA SCIENCE?

- *“Data Science is a new term. But in the same sense as Columbus was discovered NEW Continent 1000 years ago.”*

# WHAT IS DATA SCIENCE?

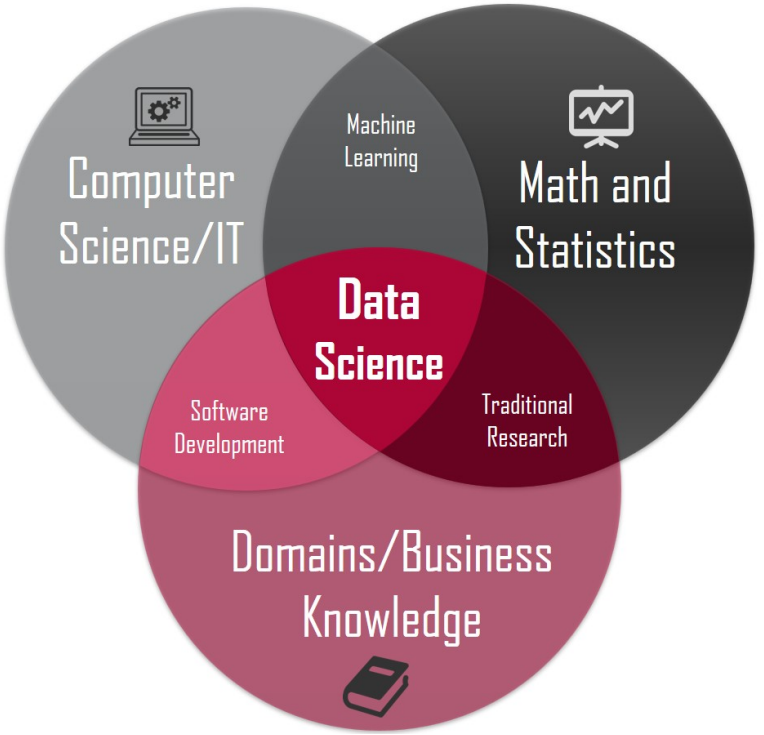
- a multi-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data.




Source: https://bit.ly/30dekJB

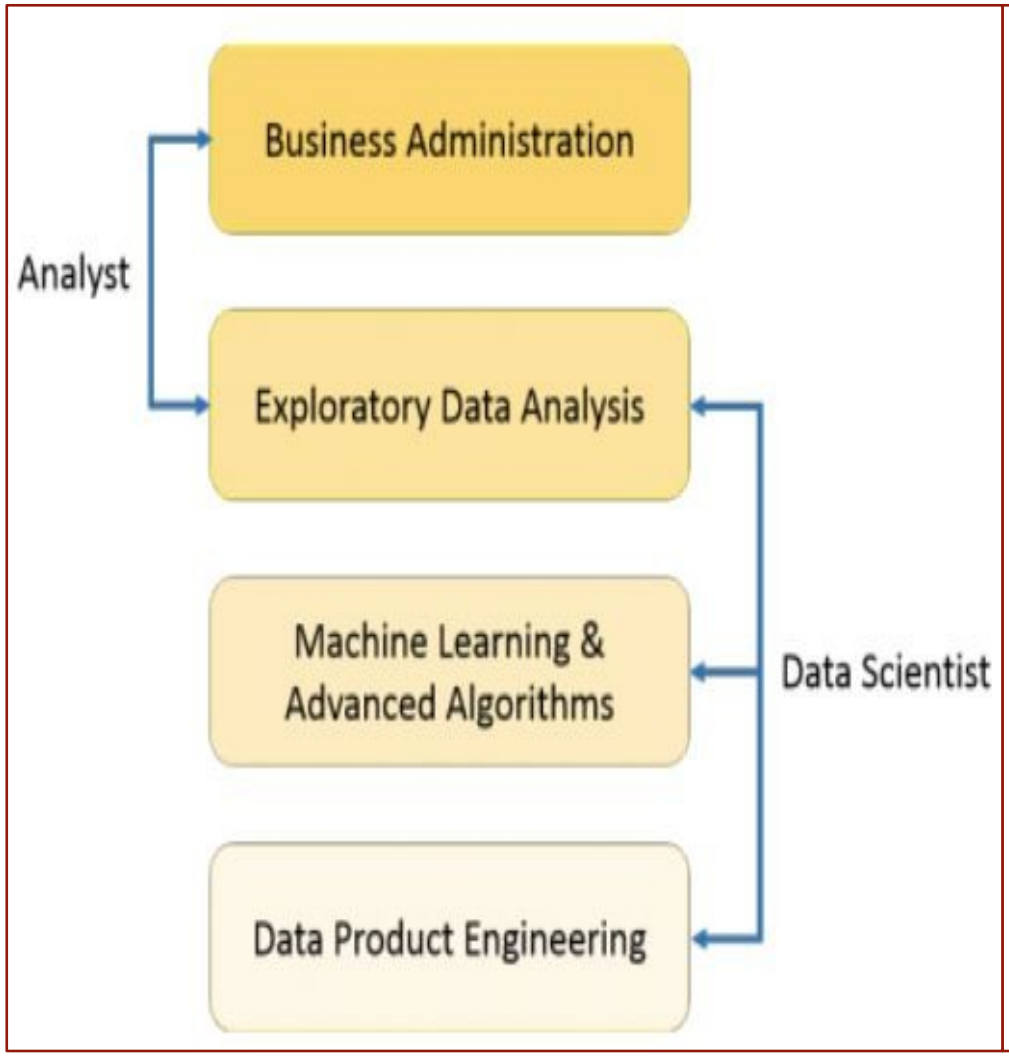
# WHAT IS DATA SCIENCE?

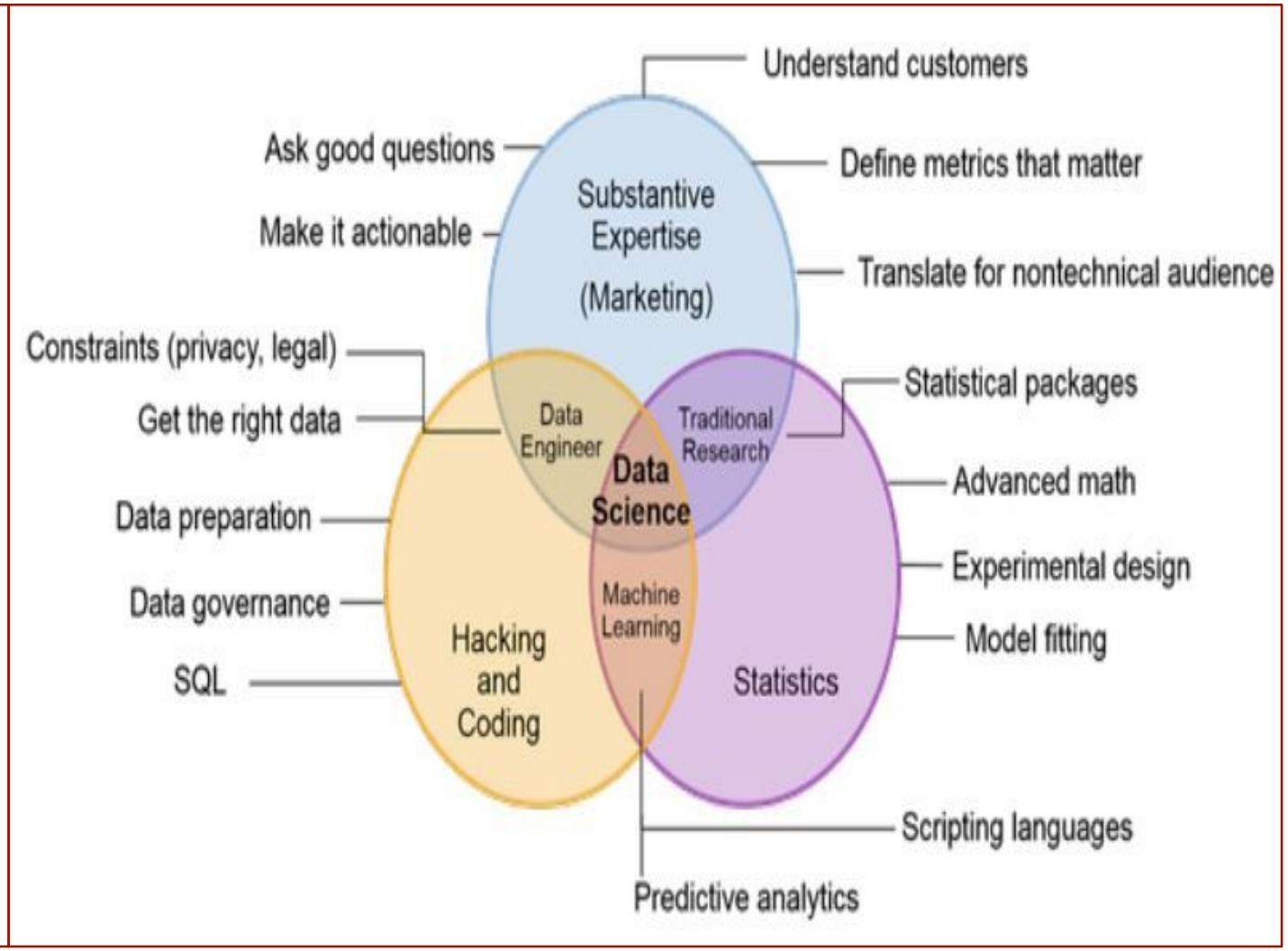
- a "concept to unify statistics, data analysis, machine learning and their related methods" in order to "understand and analyze actual phenomena" with data.
- employs techniques and theories drawn from many fields within the context of mathematics, statistics, computer science, and information science.




Source: https://bit.ly/2YTRQ3w

# WHAT IS DATA SCIENCE?

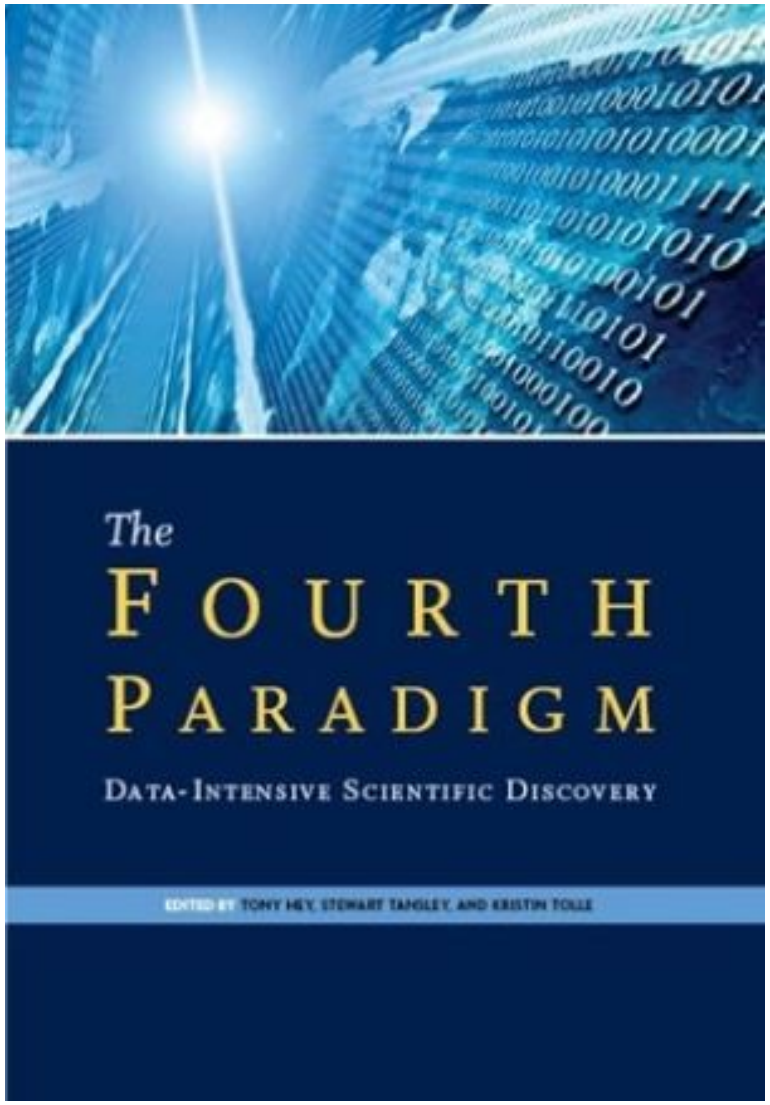


# WHAT IS DATA SCIENCE?

***Fourth Paradigm of Science***

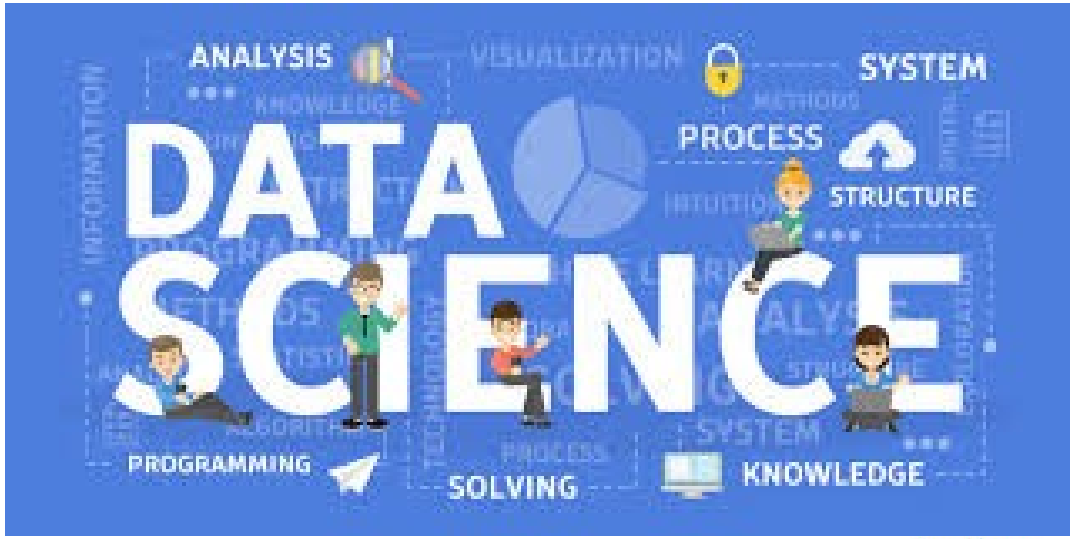
- Thousand of years
  - Empirical
- Few hundred of years
  - Theoretical
- Last fifty years
  - Computational
  - “Query the world”
- Last twenty years
  - eScience (Data Science)
  - “Download the world”

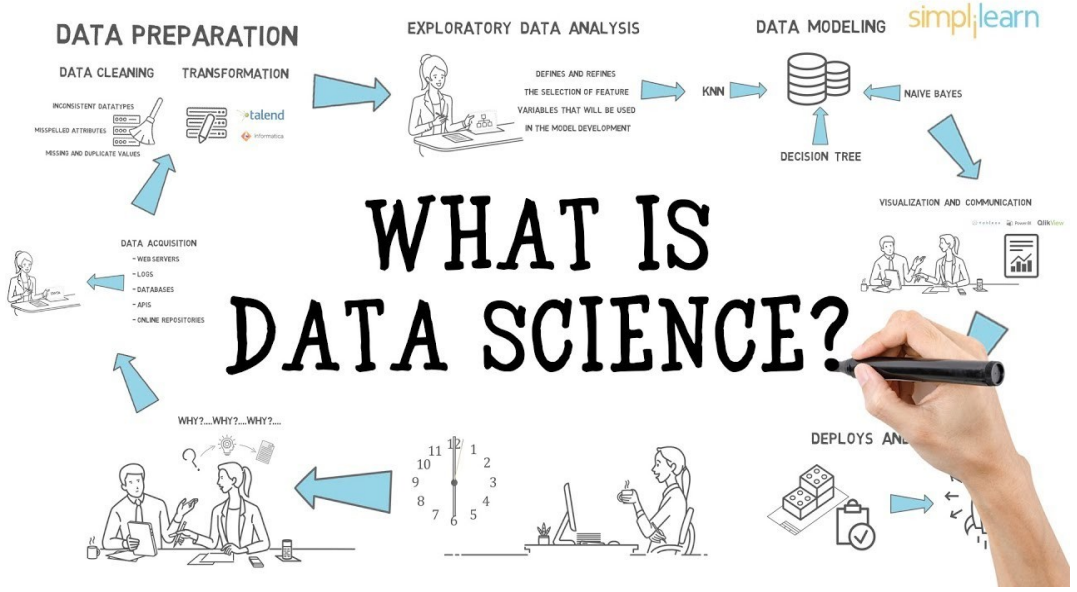


# WHAT IS DATA SCIENCE?

**Data Science and others**

- Statistics
- Big Data Analytics
- Business Analytics
- Business Intelligence
- Data(base) Management
- Visualization
- Machine Learning
- Data Mining
- Artificial Intelligence
- Predictive Modelling



# WHAT IS DATA SCIENCE?

## Big Data Science Tasks

- Facebooks
- Amazon
- Google
- LinkedIn
- Netflix
- Rozetka
- Microsoft



















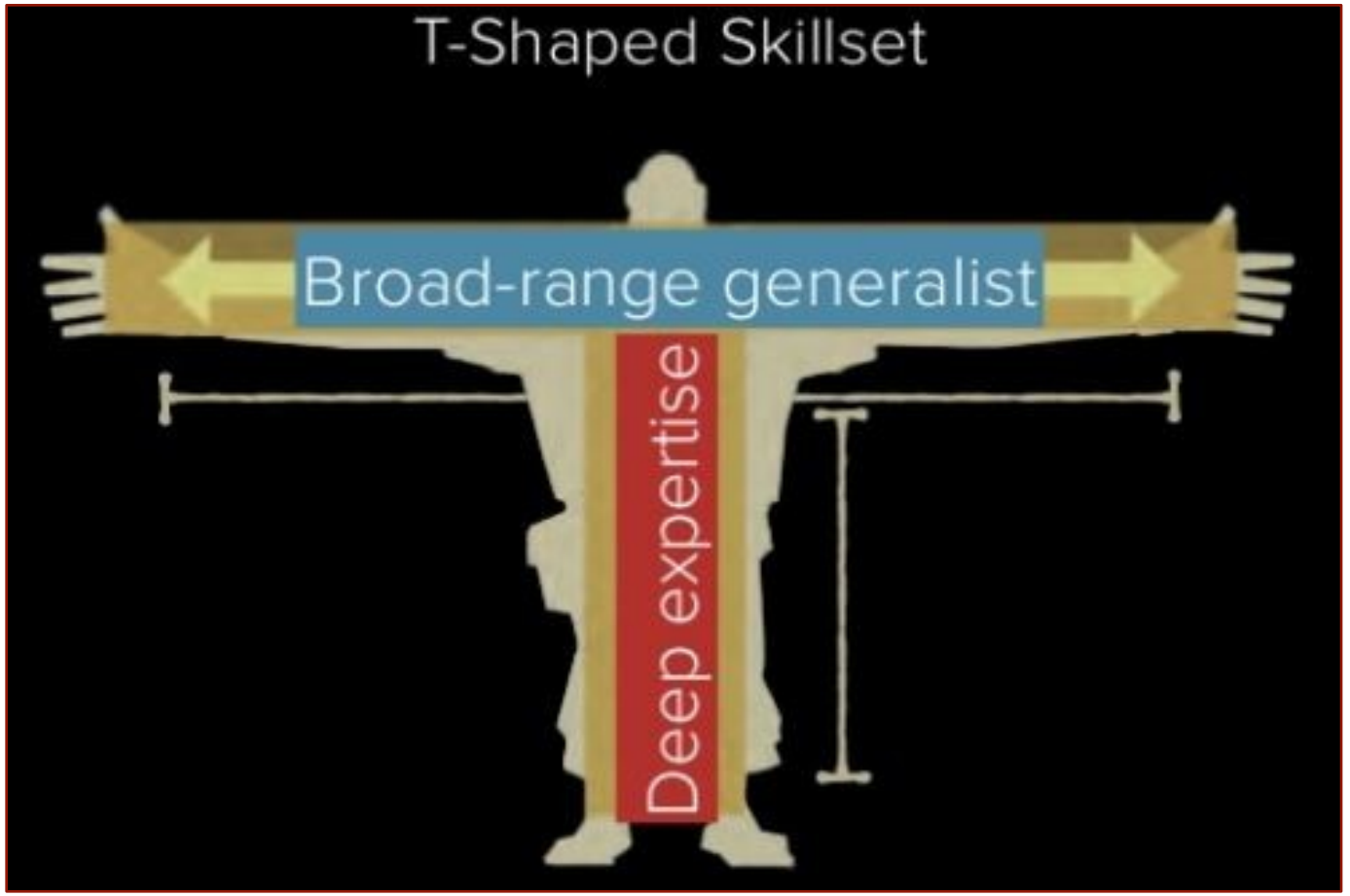

# WHAT IS DATA SCIENCE?

**Regular Data Science**

- Data Analysis
- Modelling Statistics
- Engineering / Prototyping

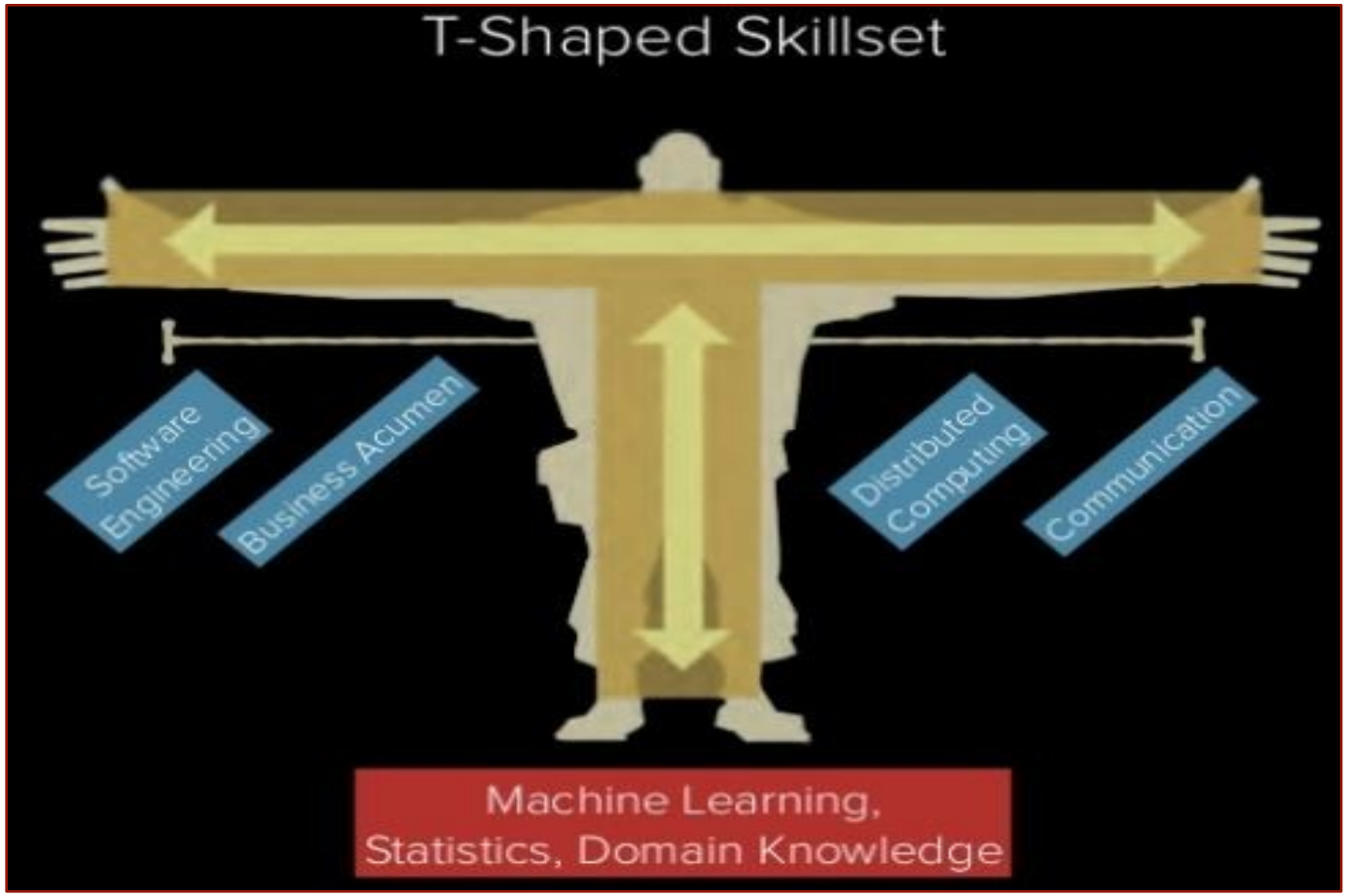
# WHAT IS DATA SCIENCE?

## What do people look for in a data scientist?



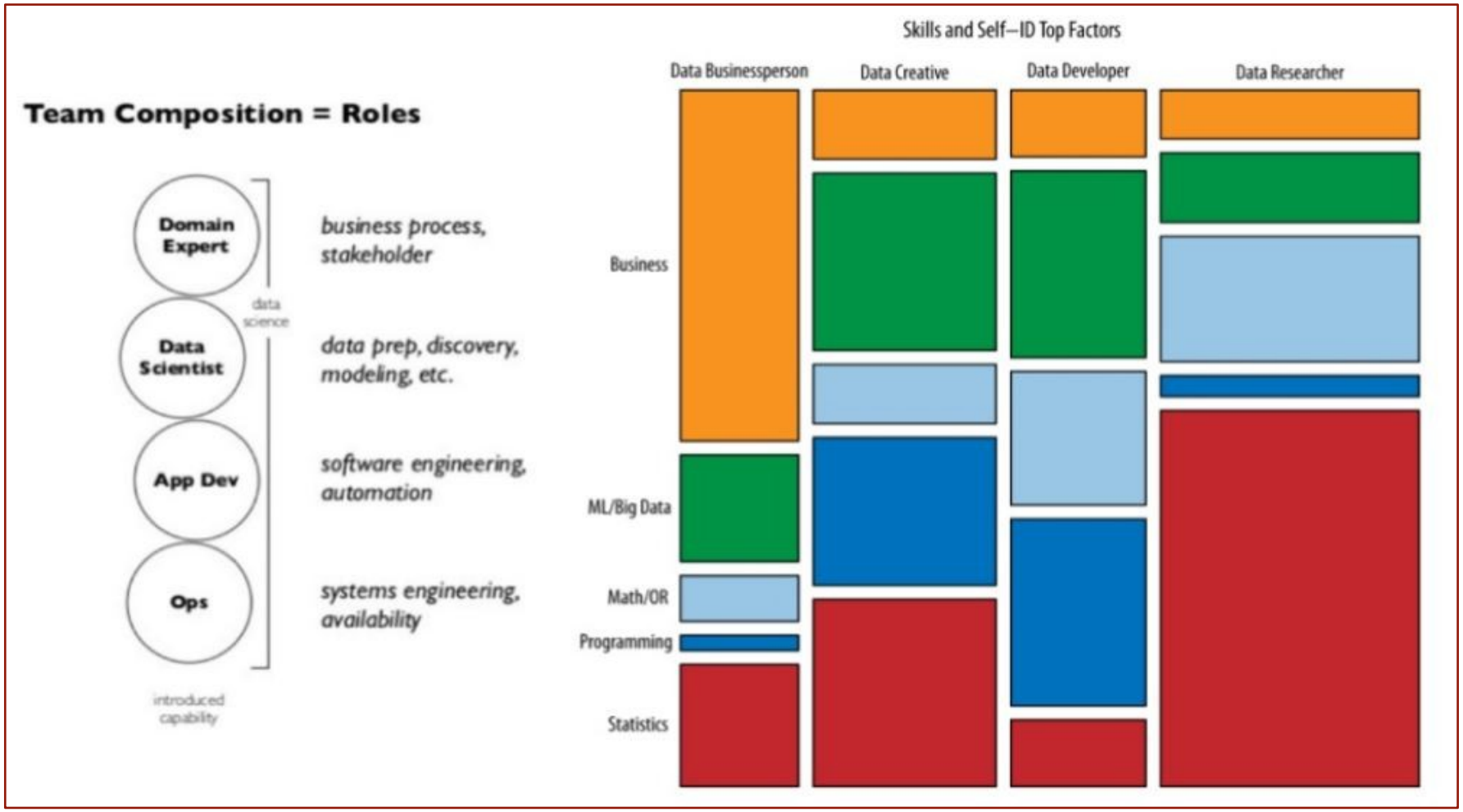
# WHAT IS DATA SCIENCE?

## What do people look for in a data scientist?



# WHAT IS DATA SCIENCE?

## Data Science Roles



# WHAT IS DATA SCIENCE?

## Roles Required in Data Science Project

**Data Scientist**

- Prove / disprove hypotheses.
- Information and Data gathering.
- Data wrangling.
- Algorithm and ML models.
- Communication.



**Data Engineer**

- Build Data Driven Platforms.
- Operationalize Algorithms and Machine Learning models.
- Data Integration.



**Visualization Expert**

- Storytelling.
- Build Dashboards and other Data visualizations.
- Provide insight through visual means.



**Process Owner**

- Project Management.
- Manage stakeholder expectations.
- Maintain a Vision.
- Facilitate.



Source: https://bit.ly/2z5sYqf

# WHAT IS DATA SCIENCE?

**How to become a data scientist?**

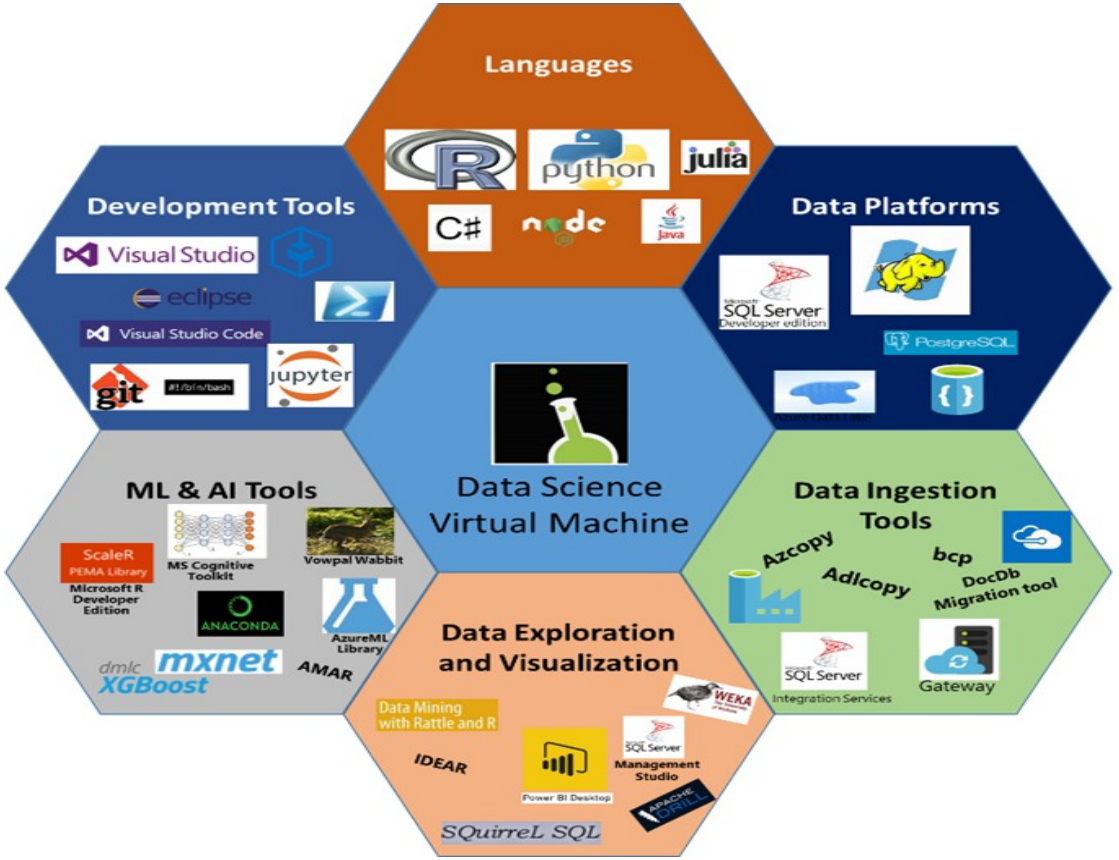
- Data Scientists need to know how to "**CODE**"



# WHAT IS DATA SCIENCE?

## How to become a data scientist?

- Other languages, tools, platforms and visualization



# WHAT IS DATA SCIENCE?

## Learning Data Science with Python - Libraries




NumPy is a library for the Python programming language, adding support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays.




A free software machine learning library that features various classification, regression and clustering algorithms including support vector machines, random forests, gradient boosting, and k-means and is designed to interoperate with the Python numerical and scientific libraries NumPy and SciPy.



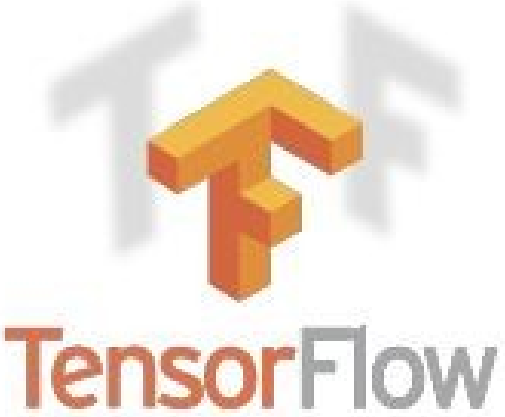

Pandas is a software library written for the Python programming language for data manipulation and analysis. In particular, it offers data structures and operations for manipulating numerical tables and time series.

# WHAT IS DATA SCIENCE?

## Learning Data Science with Python - Libraries




A plotting library for the Python programming language and its numerical mathematics extension NumPy




TensorFlow is an open-source software library for dataflow programming across a range of tasks. It is a symbolic math library, and is also used for machine learning applications such as neural networks.




Keras is an open source neural network library written in Python. It is capable of running on top of TensorFlow, Microsoft Cognitive Toolkit, Theano, or MXNet. It was developed with a focus on enabling fast experimentation

# WHAT IS DATA SCIENCE?

## Learning Data Science with Python - Tools




Open-source web application that allows you to create and share documents that contain live code, equations, visualizations and narrative text
http://jupyter.org/



Similar to Jupyter Notebook, but with the added benefit of "google doc" type sharing and collaboration
https://colab.research.google.com

Crestle

Effortless infrastructure for deep learning

Crestle is your GPU-enabled Jupyter environment in the cloud.
https://www.crestle.com/

# WHAT IS DATA SCIENCE?

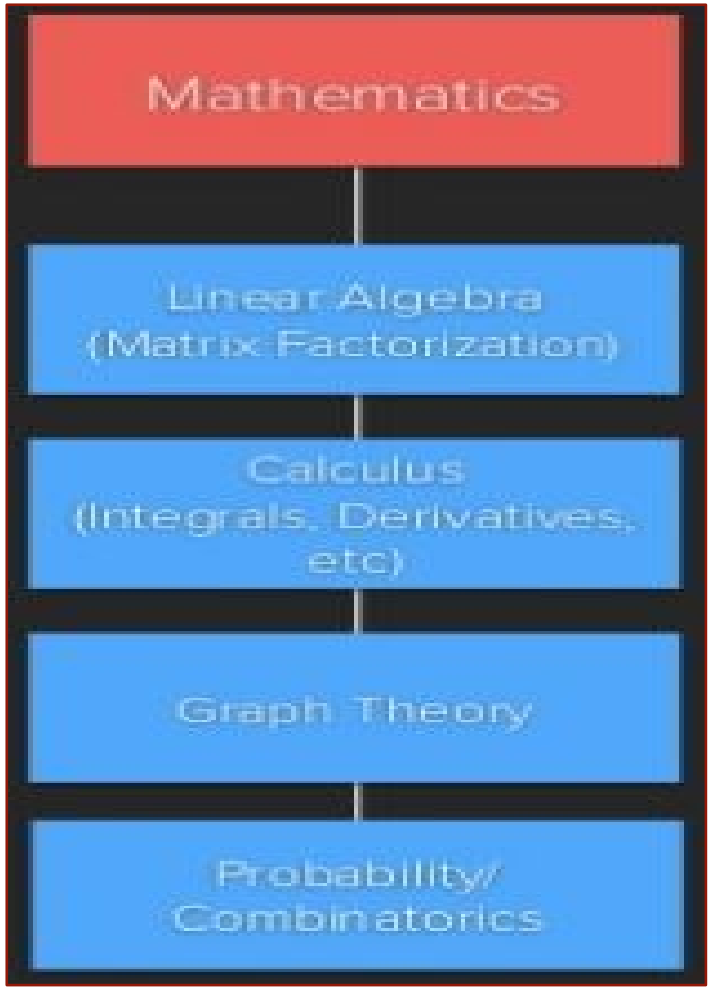
**How to become a data scientist?**

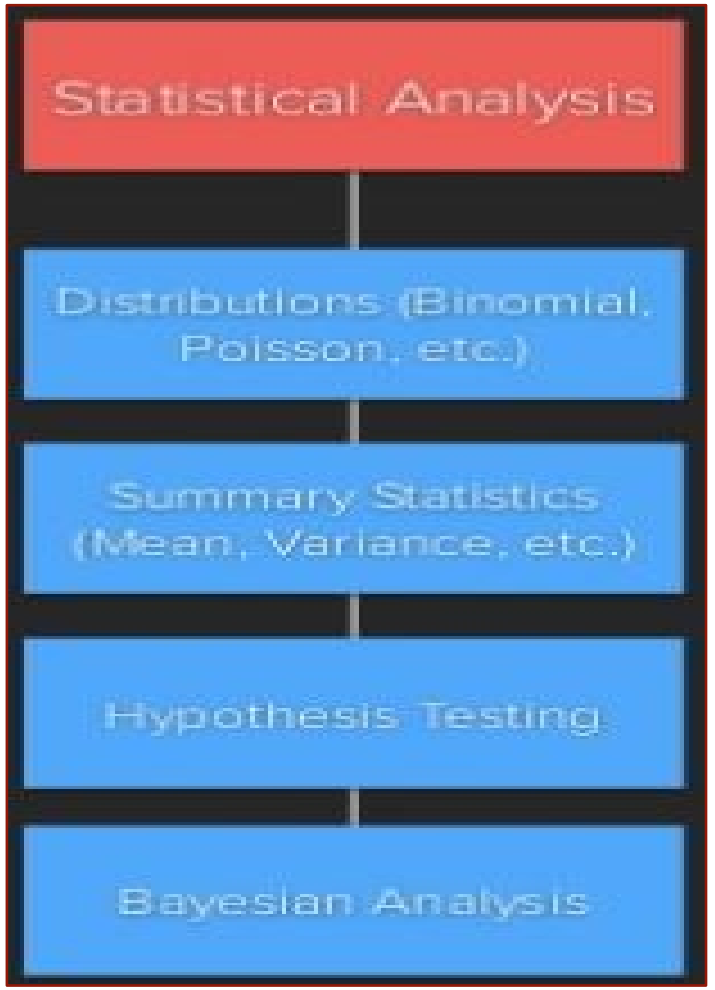
- Learn to code



# WHAT IS DATA SCIENCE?

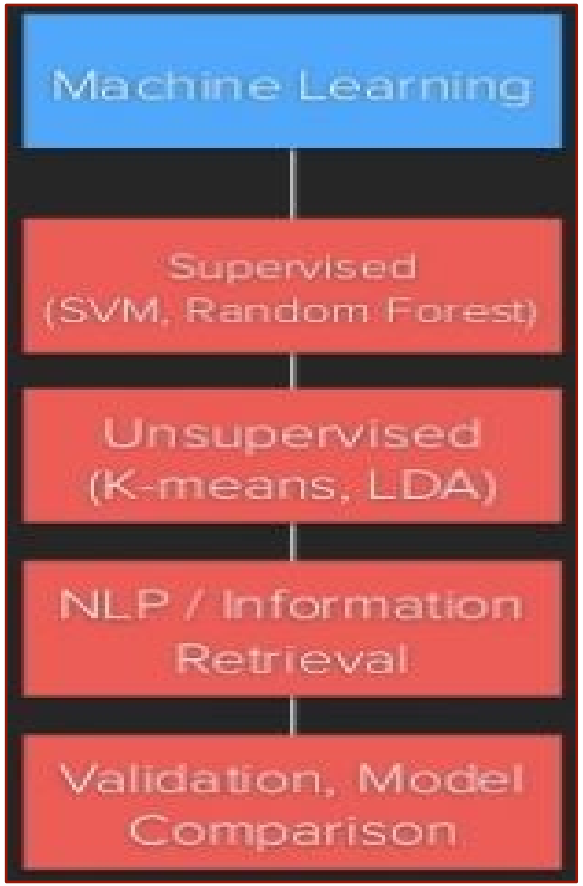
## Data Scientist need to comfortable with:

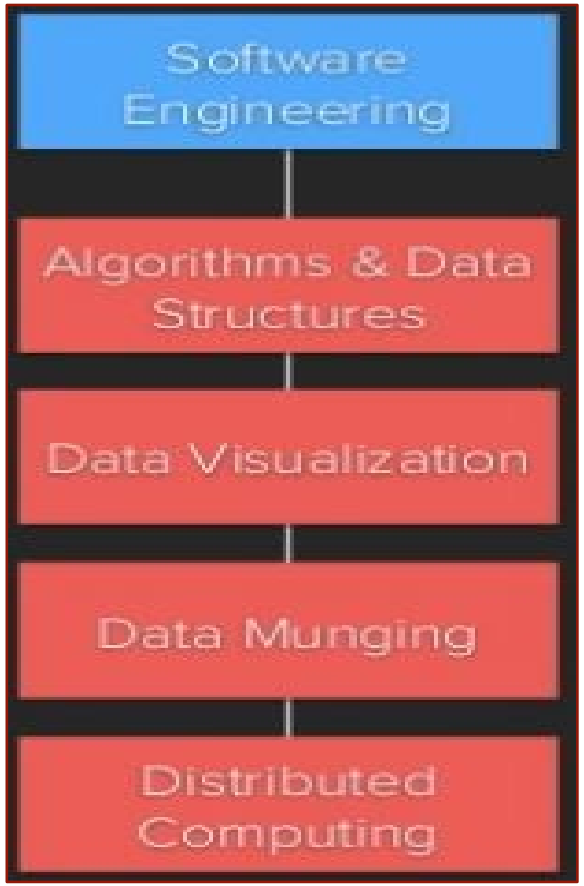


# WHAT IS DATA SCIENCE?

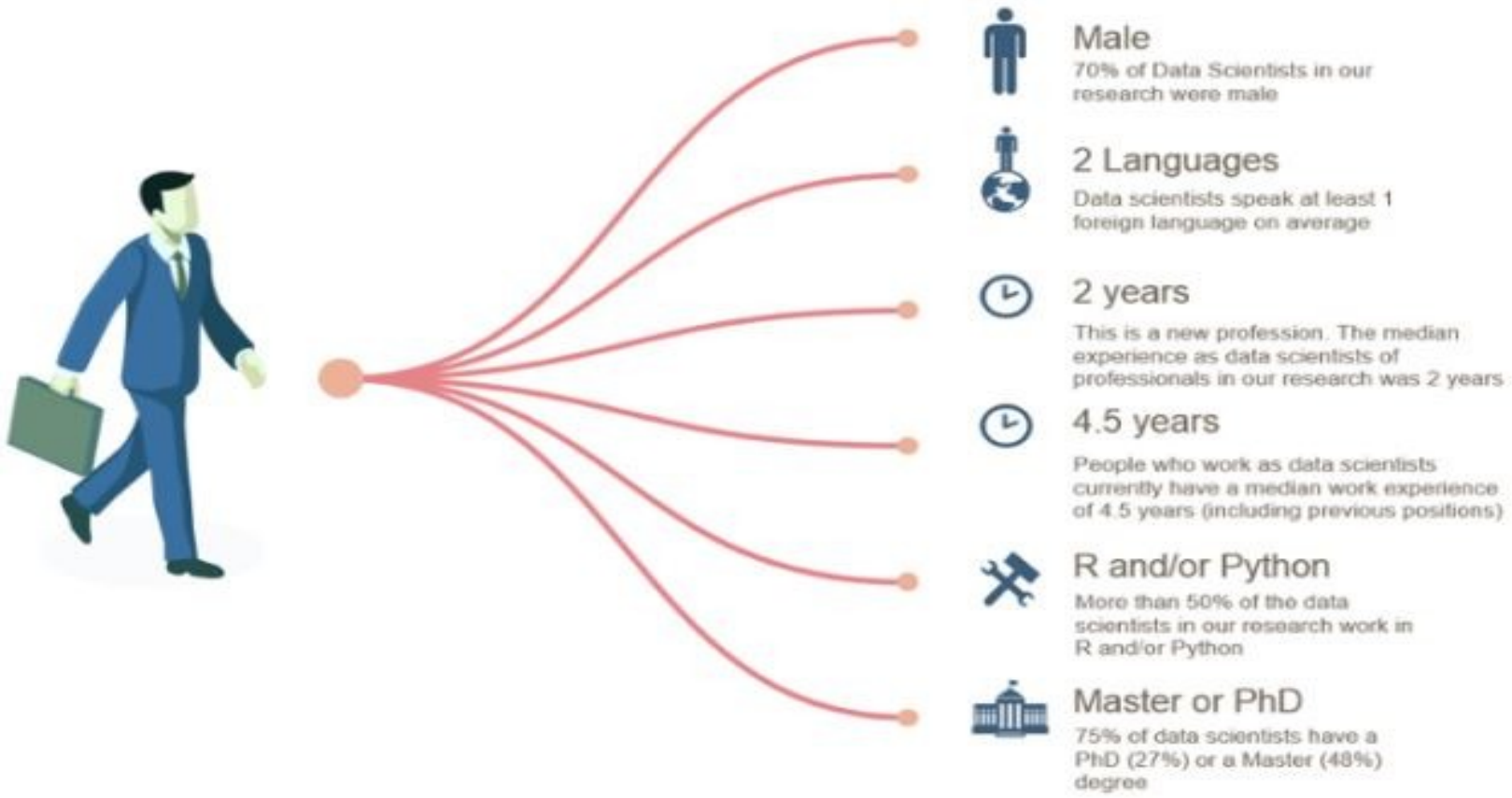
**Data Scientist need to learning machine learning & software engineering**



# WHAT IS DATA SCIENCE?

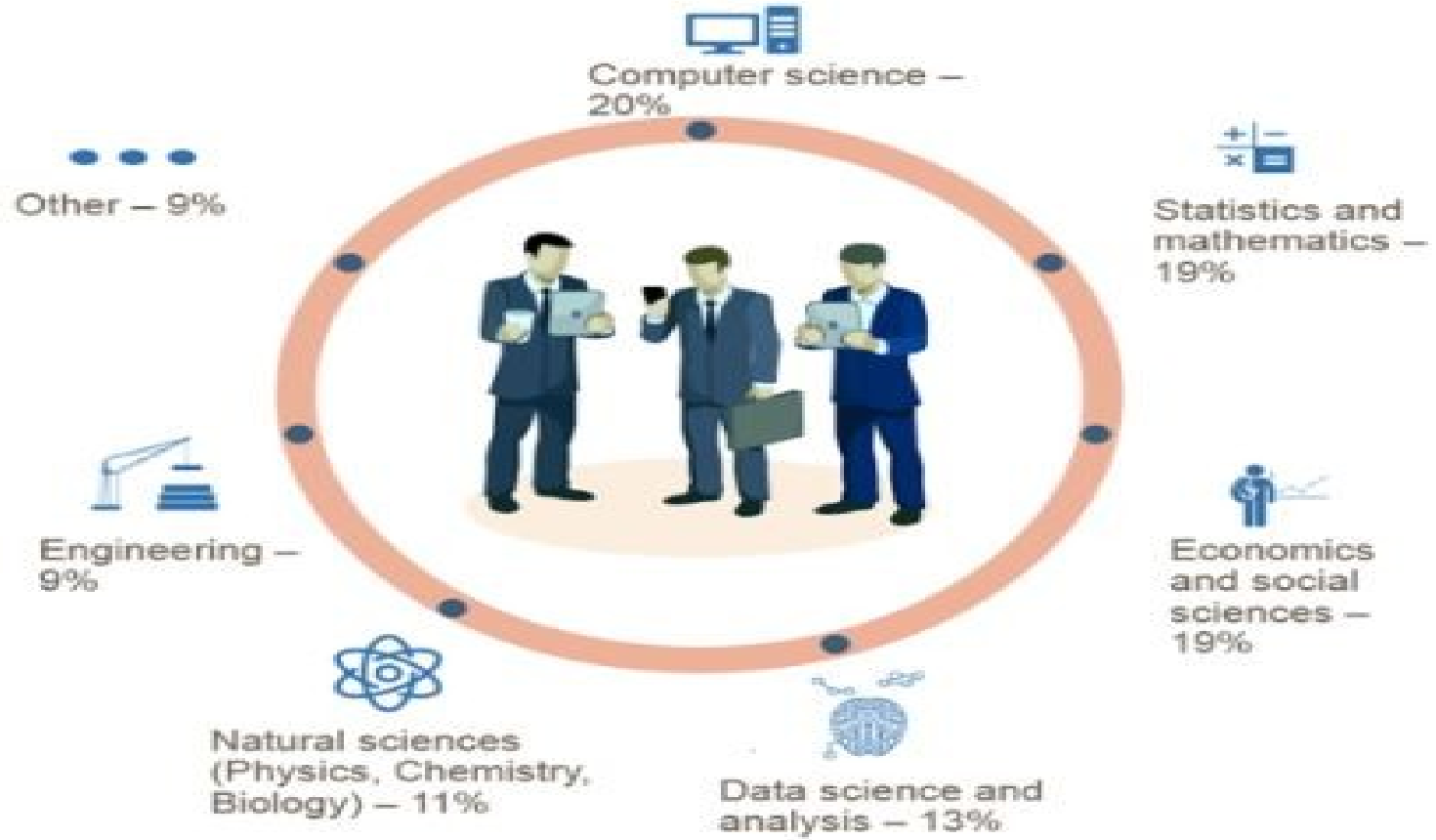
## Who are the Data Scientist?




https://www.datascience.com/blog/data-scientist-skills

# WHAT IS DATA SCIENCE?

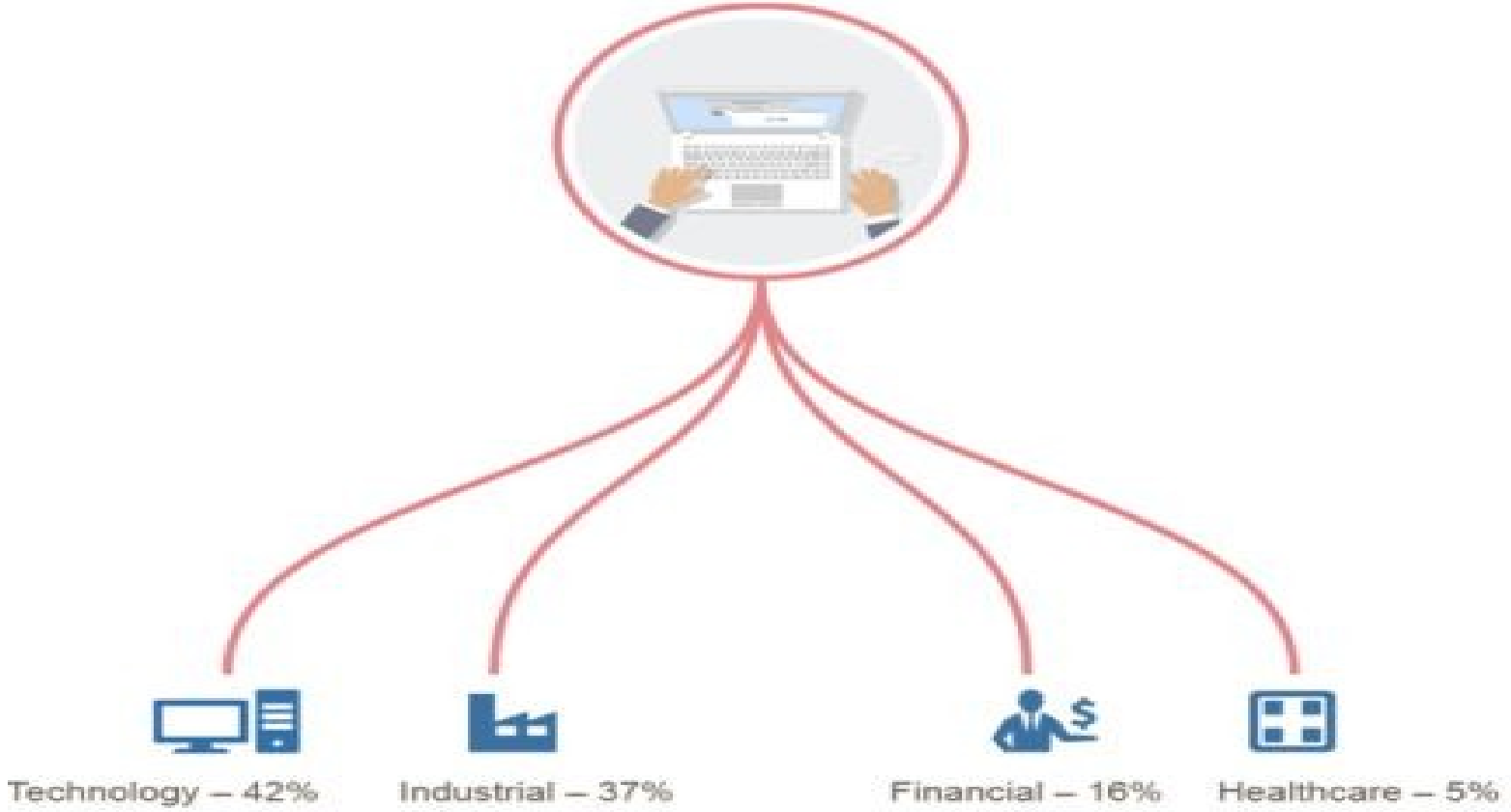
## Who are the Data Scientist?




https://www.datascience.com/blog/data-scientist-skills

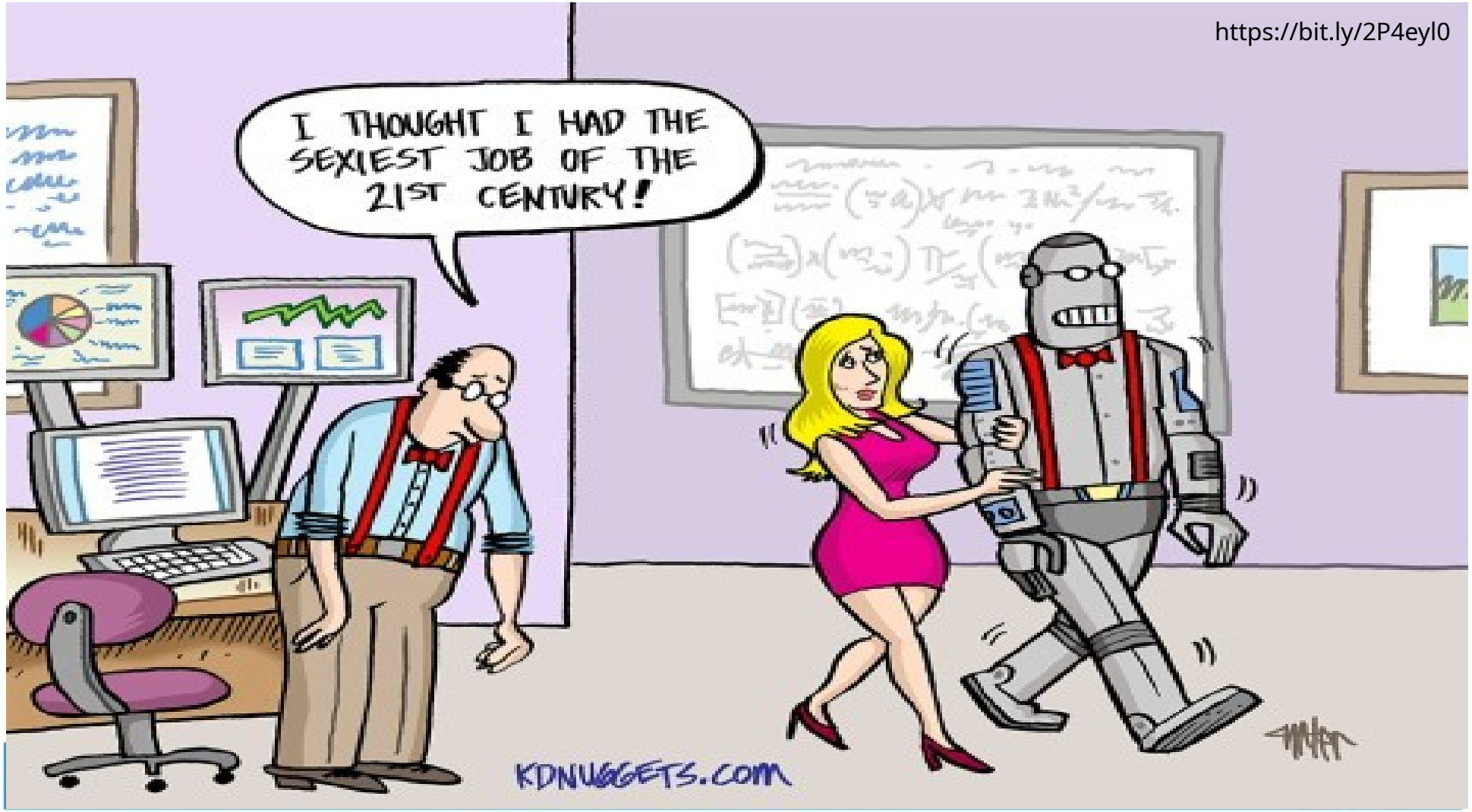
# WHAT IS DATA SCIENCE?

## Who are the Data Scientist?




https://www.datascience.com/blog/data-scientist-skills

# WHAT IS DATA SCIENCE?



# WHAT IS DATA SCIENCE?

## Data Scientist Salaries in United States

Very High Confidence

4,354 Salaries  Updated Aug 17, 2019

Industries | Company Sizes | Years of Experience

Average Base Pay

**$117,345 /yr**




Additional Cash Compensation

| | |
|---|---|
| Average | $11,530 |
| Range | $3,933 - $26,784 |

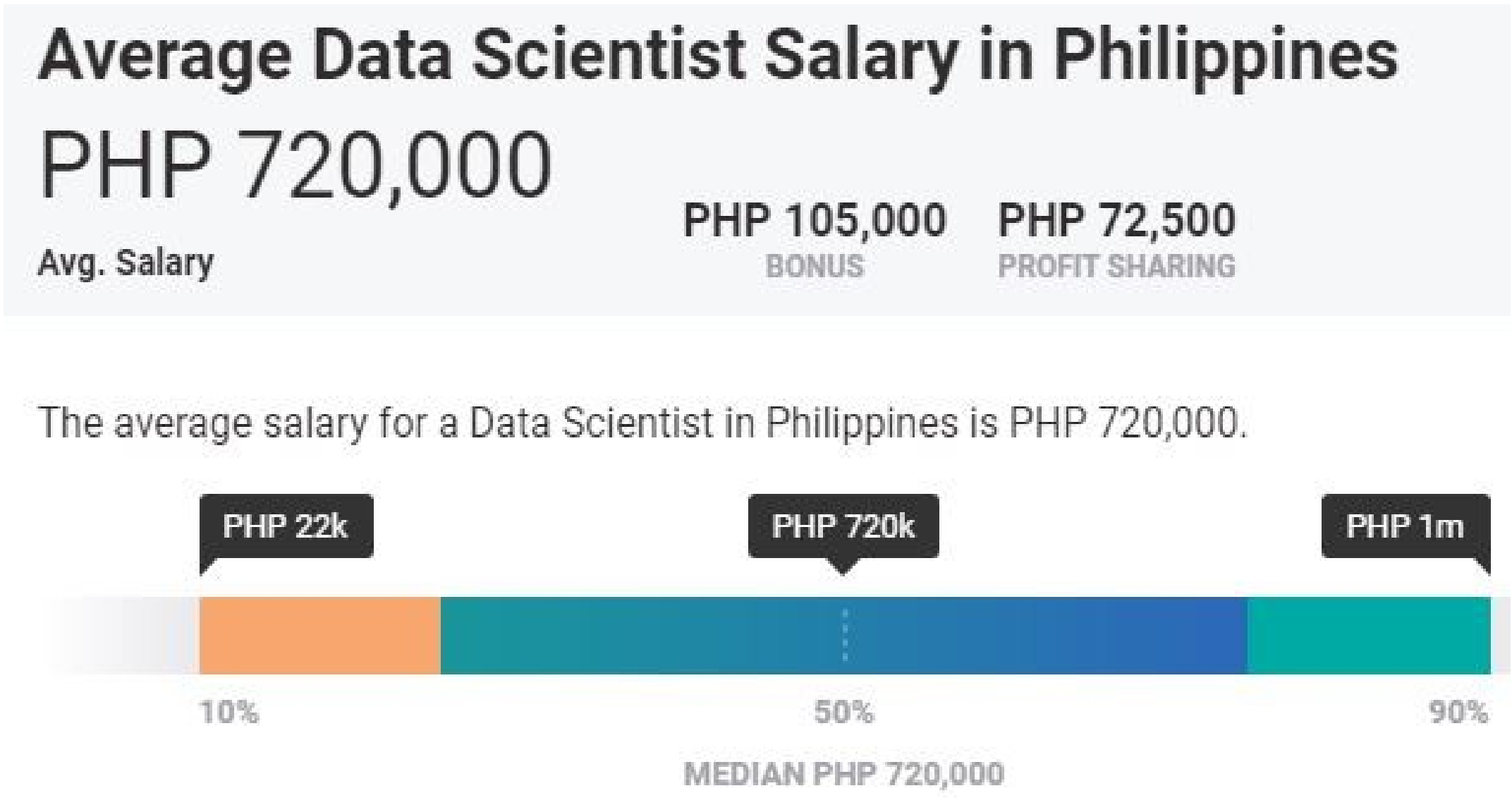
**How much does a Data Scientist make?**

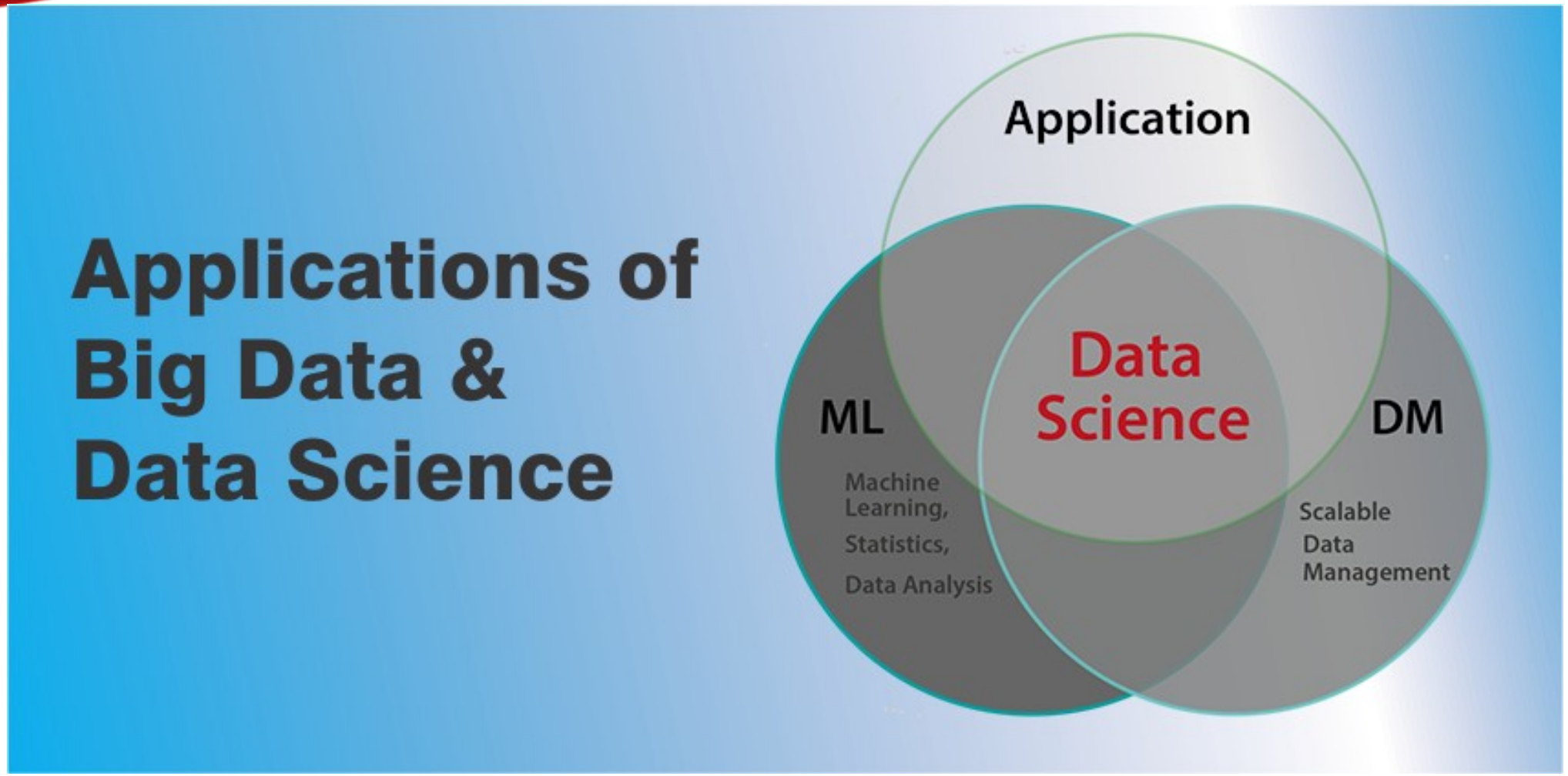
The national average salary for a Data Scientist is $117,345 in United States. Filter by location to see Data Scientist salaries in your area. Salary estimates are based on 4,354 salaries submitted anonymously to Glassdoor by Data Scientist employees.

Salaries for Related Job Titles

| | |
|---|---|
| Data Analyst | $67K |
| Data Scientist Intern | $96K |
| Quantitative Analyst | $116K |
| Senior Data Scientist | $137K |

# WHAT IS DATA SCIENCE?






# APPLICATION OF DATA SCIENCE

# APPLICATIONS OF DATA SCIENCE

- Security



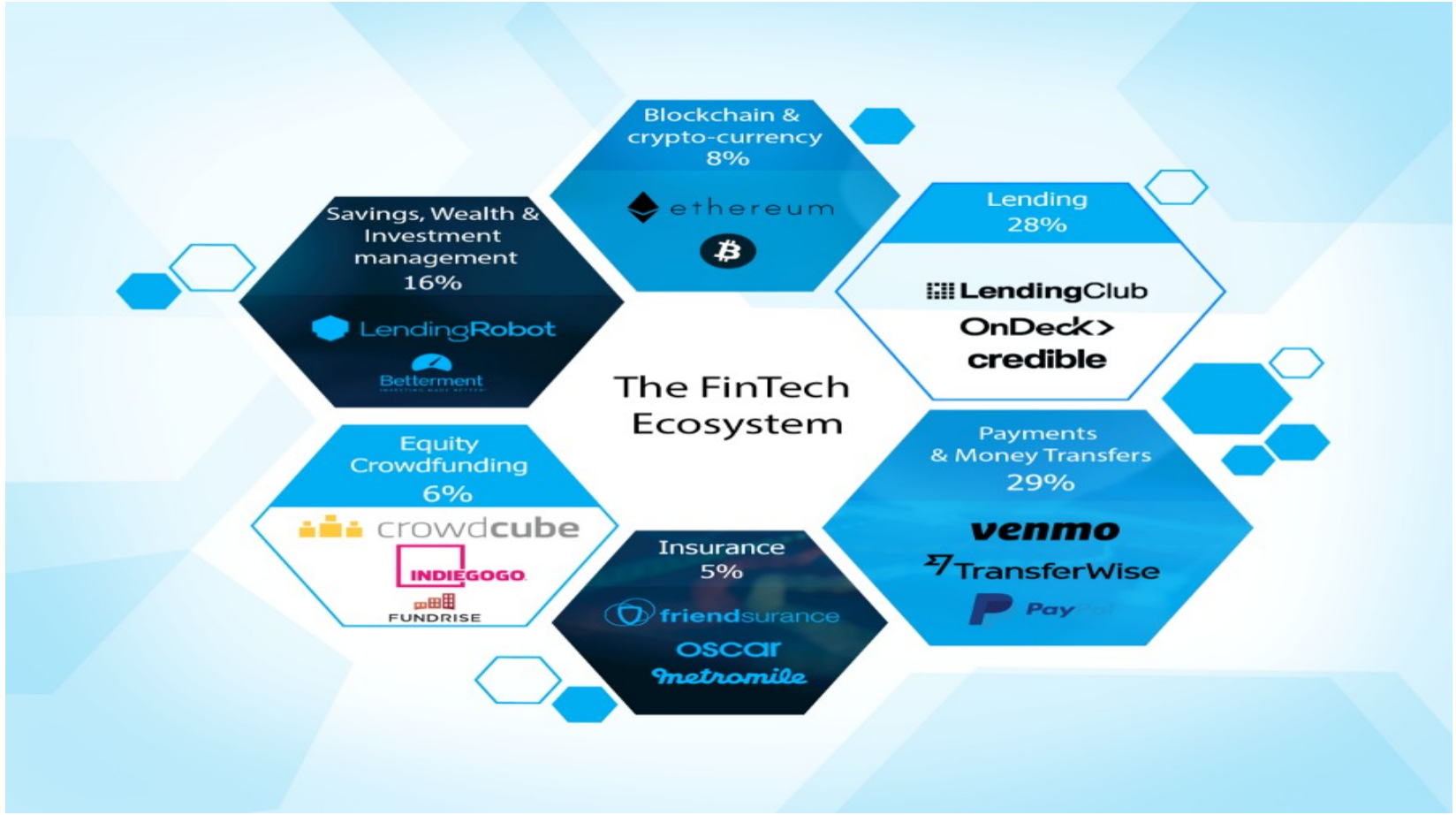
# APPLICATIONS OF DATA SCIENCE

- Sports



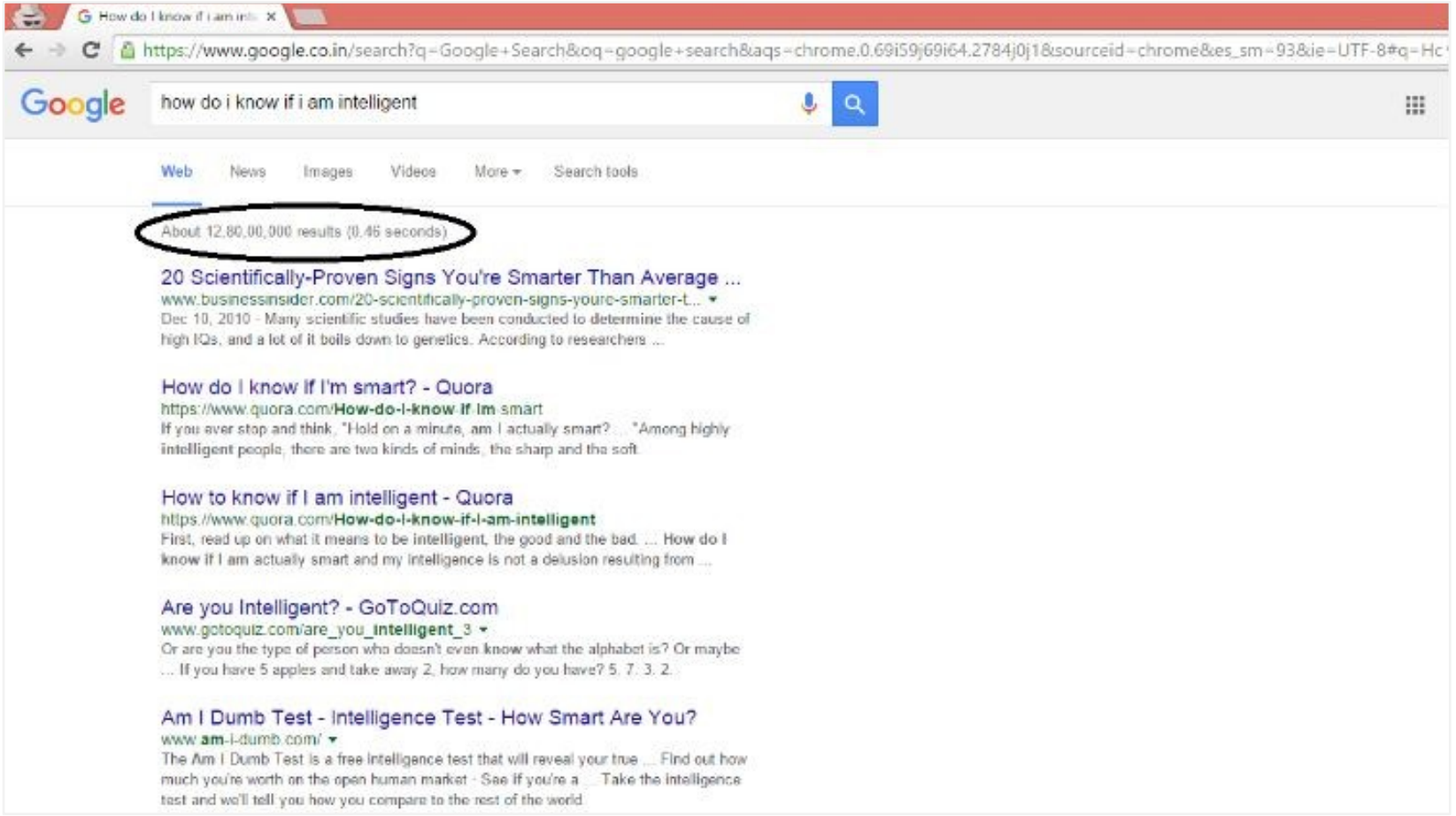
# APPLICATIONS OF DATA SCIENCE

- Banking and Finance



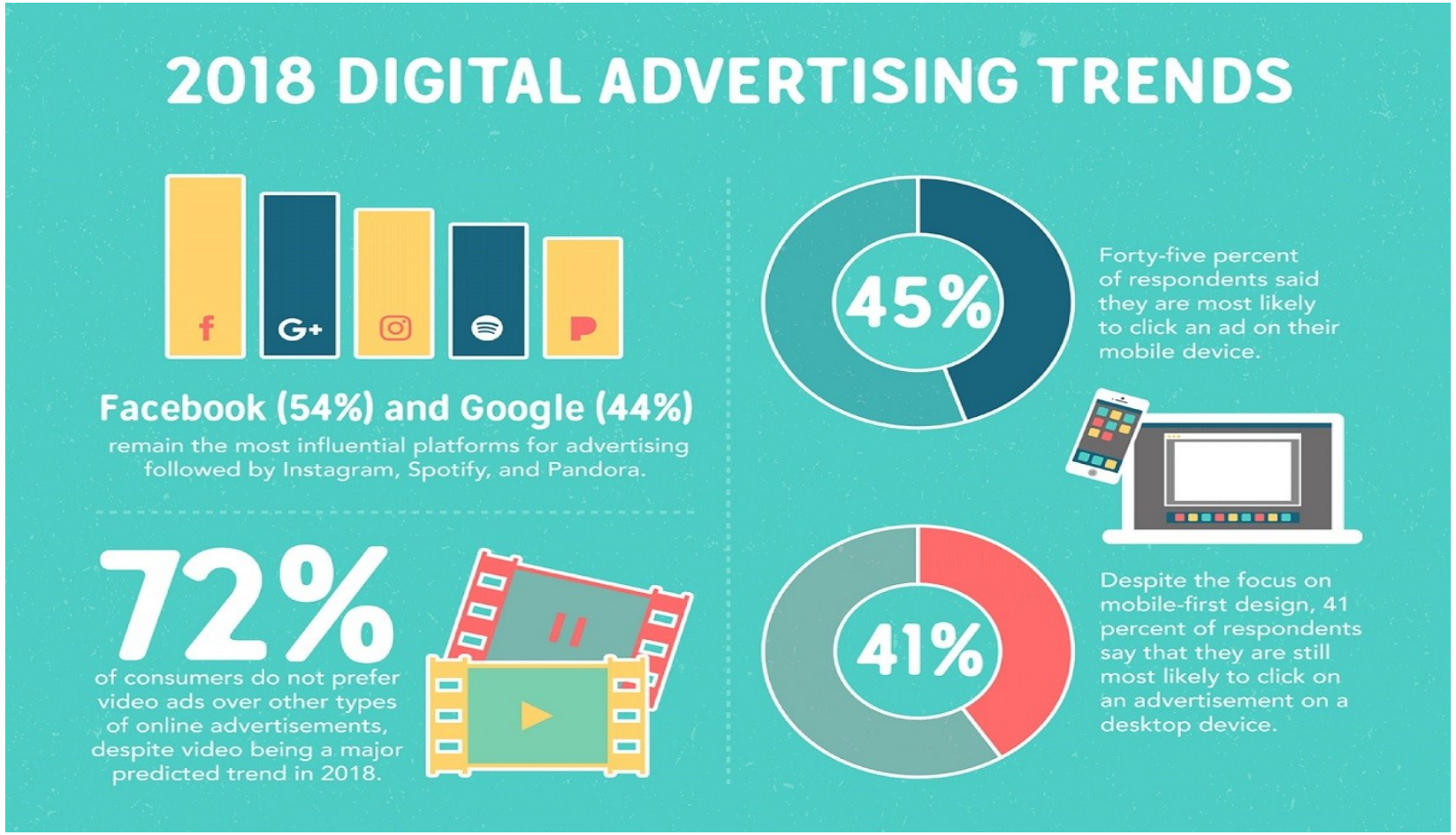
# APPLICATIONS OF DATA SCIENCE

- Internet Search



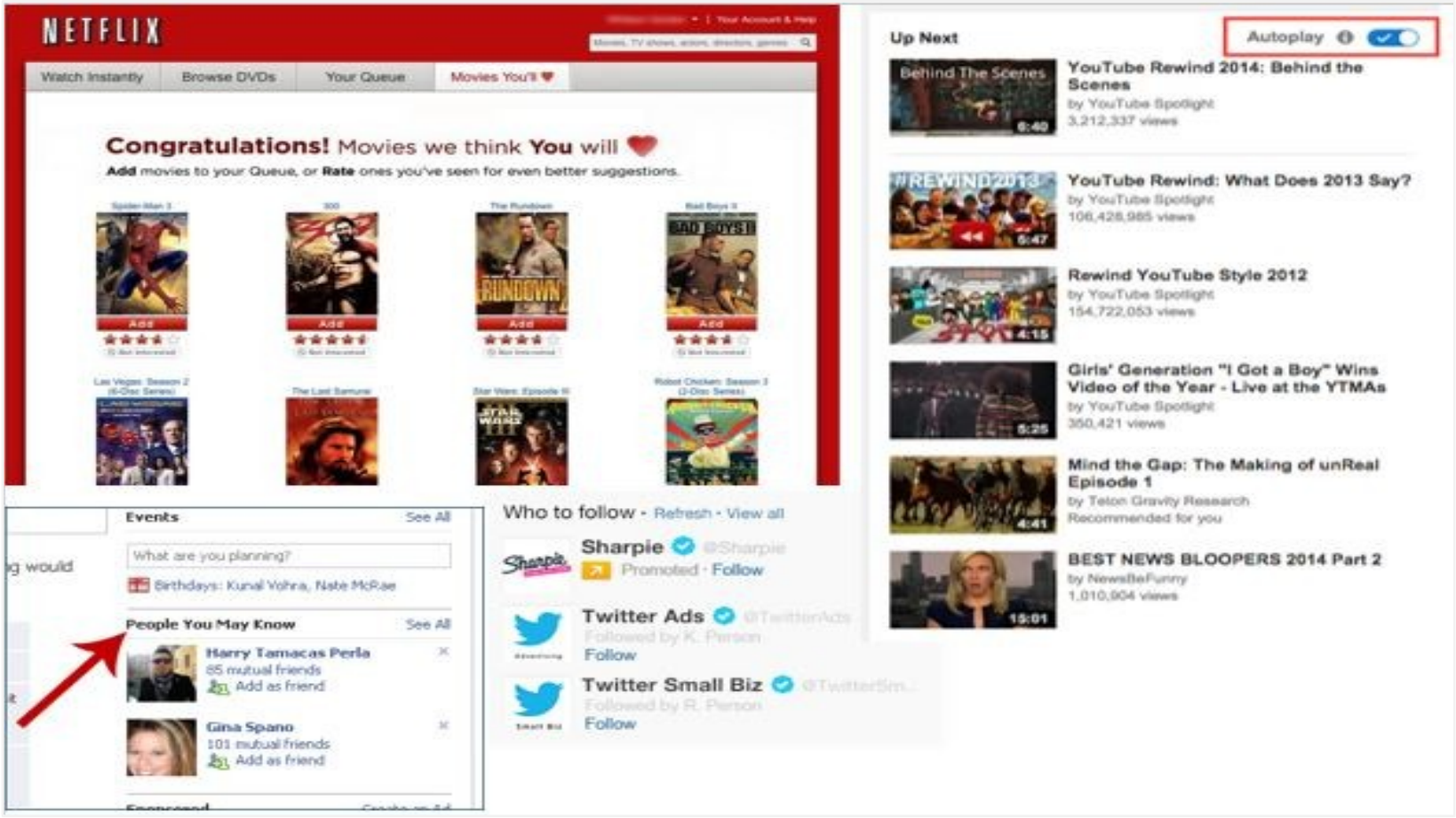
# APPLICATIONS OF DATA SCIENCE

- Digital Advertisements



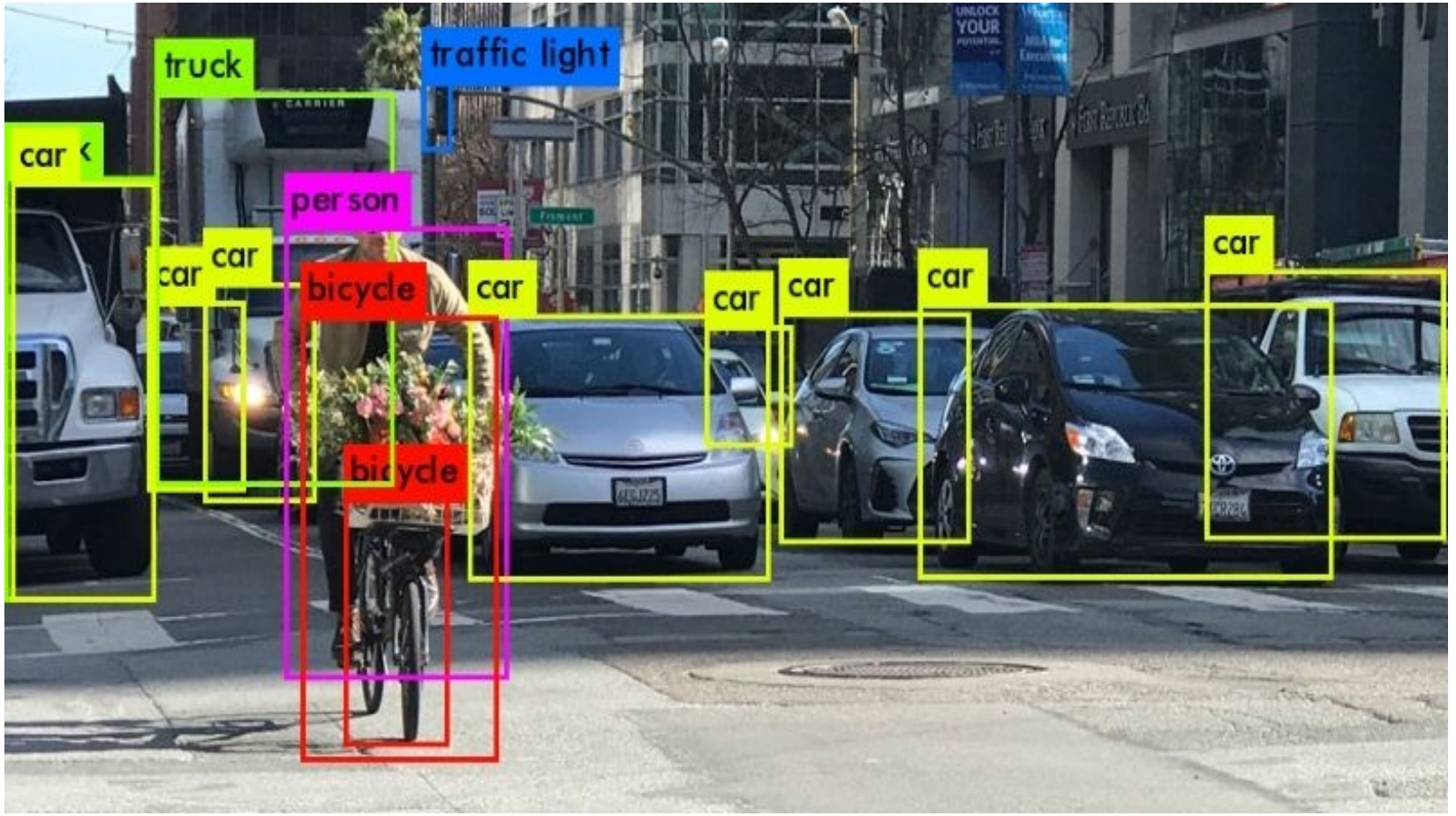
# APPLICATIONS OF DATA SCIENCE

- Recommender System



# APPLICATIONS OF DATA SCIENCE

- Image Processing



# APPLICATIONS OF DATA SCIENCE

- Speech Recognition



# APPLICATIONS OF DATA SCIENCE

- Gaming



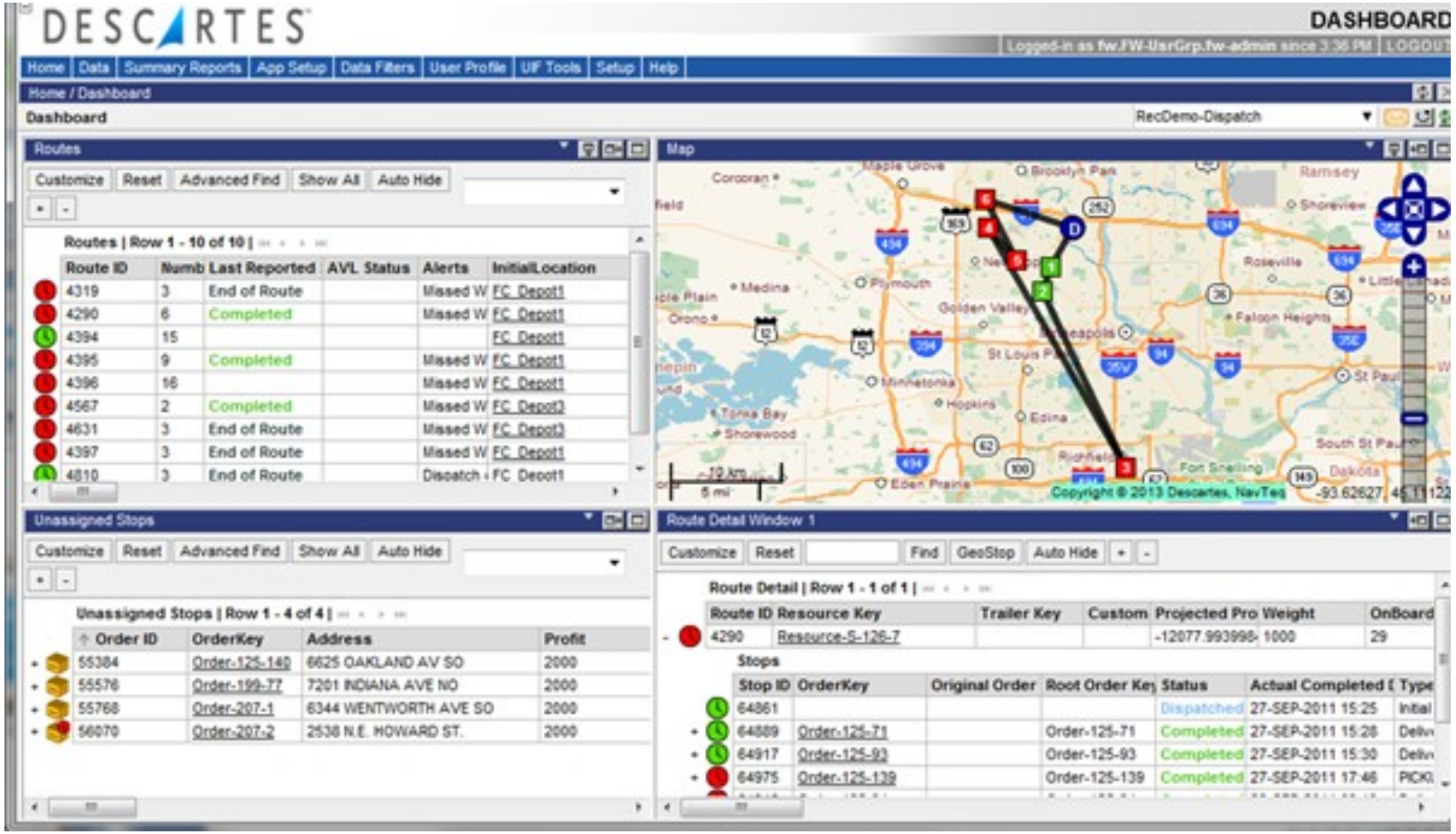
# APPLICATIONS OF DATA SCIENCE

- Price Comparison Websites



# APPLICATIONS OF DATA SCIENCE

- Airline Routing Planning



# APPLICATIONS OF DATA SCIENCE

- Fraud and Risk Detection



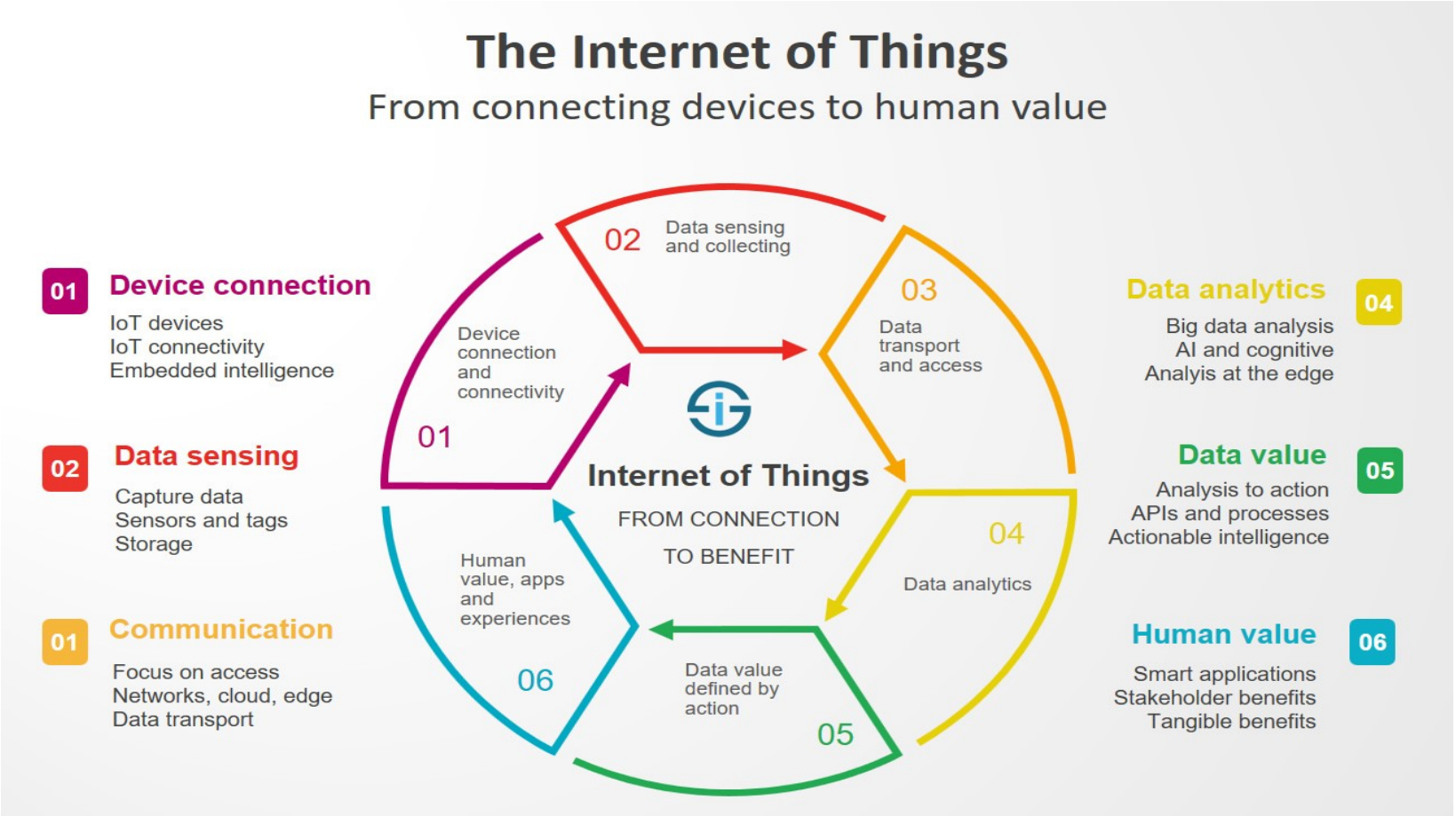
# APPLICATIONS OF DATA SCIENCE

- Delivery Logistics



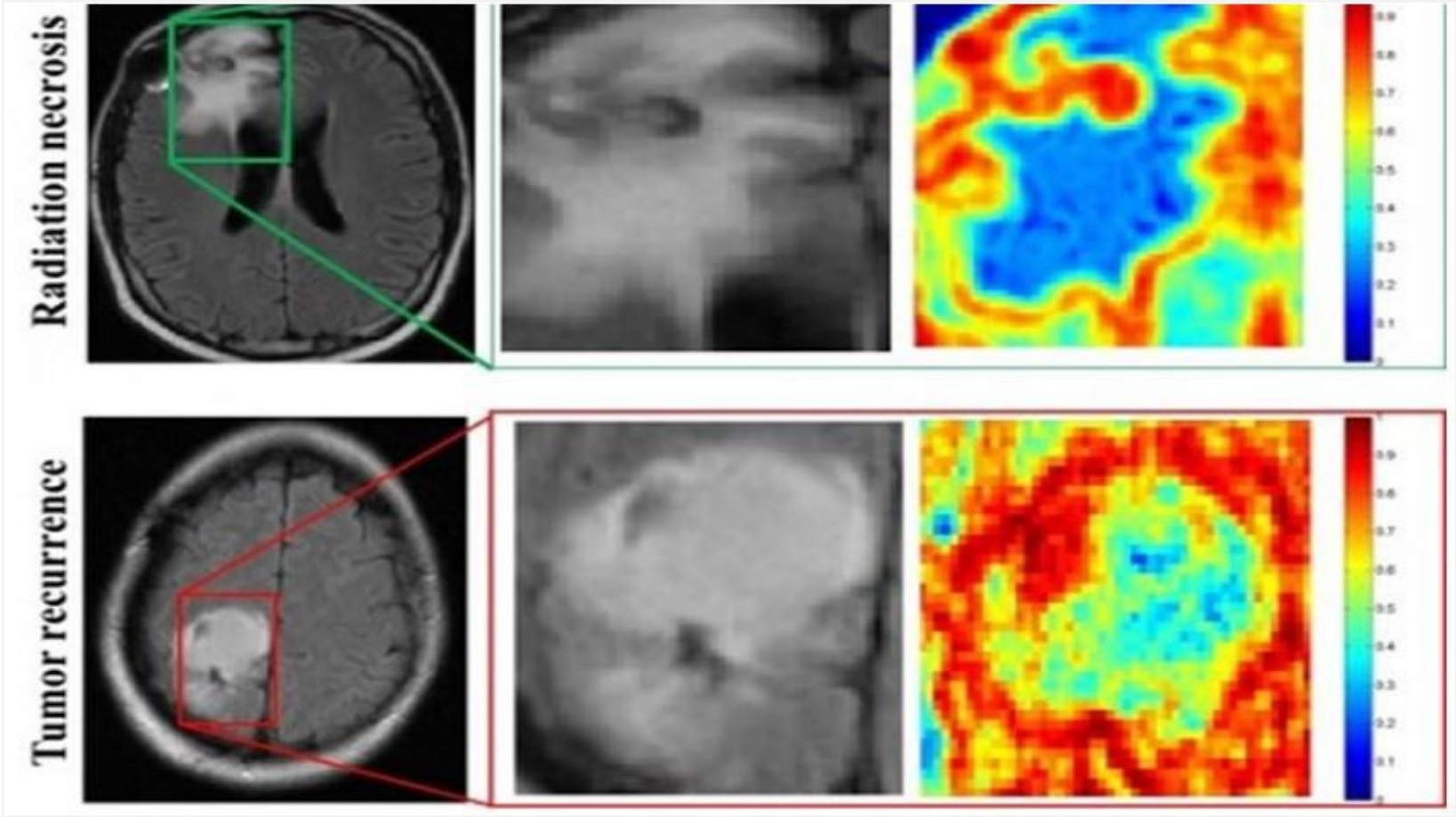
# APPLICATIONS OF DATA SCIENCE

- Internet of Things (IoT)



# APPLICATIONS OF DATA SCIENCE

- Health Care



# APPLICATIONS OF DATA SCIENCE

- Augmented Reality



# APPLICATIONS OF DATA SCIENCE

- Self-Driving Cars



# APPLICATIONS OF DATA SCIENCE

- Robots






# IMPACT OF DATA SCIENCE ON SOCIETY

# IMPACT OF DATA SCIENCE ON SOCIETY

- Saving Energy



# IMPACT OF DATA SCIENCE ON SOCIETY

- Data-Driven Hospitals



# IMPACT OF DATA SCIENCE ON SOCIETY

- A Cleaner Environment



# IMPACT OF DATA SCIENCE ON SOCIETY

- Volunteer with a socially-oriented data science program/organization



# IMPACT OF DATA SCIENCE ON SOCIETY

- Contribute via competitions



# IMPACT OF DATA SCIENCE ON SOCIETY

- Consider solutions to real-world problems that you encounter




Sample data produced by Gap Minder Foundation, demonstrating the relationship between life expectancy and country's income level

# IMPACT OF DATA SCIENCE ON SOCIETY

- Be thoughtful in professional work






# IMPORTANCE OF DATA SCIENCE

# IMPORTANCE OF DATA SCIENCE

1. Data science helps brands to understand their customers in a much enhanced and empowered manner.

2. It allows brands to communicate their story in such a engaging and powerful manner.

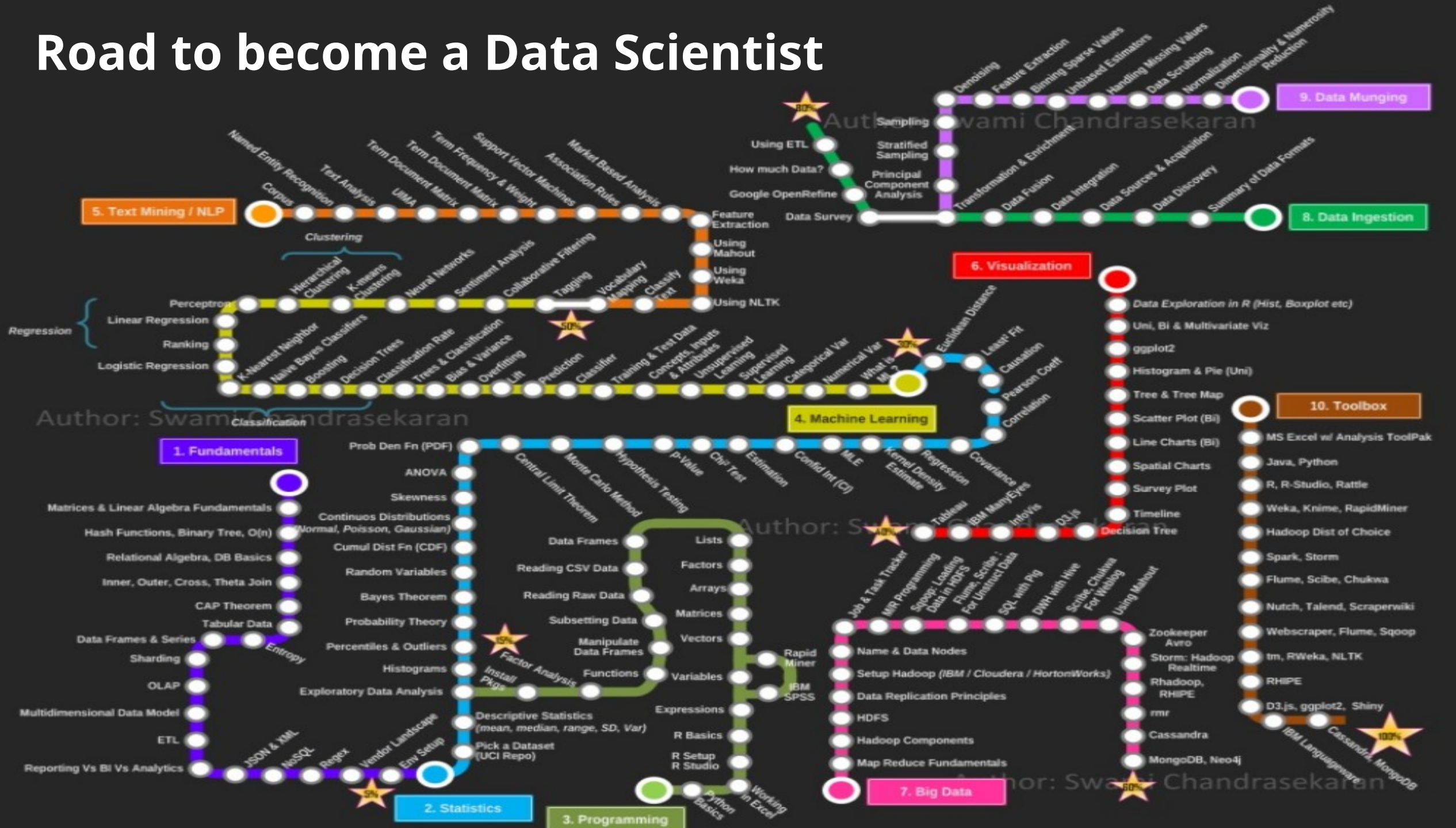
3. Big Data is a new field that is constantly growing and evolving.

# IMPORTANCE OF DATA SCIENCE

4. Its findings and results can be applied to almost any sector like travel, healthcare and education among others.

5. Data science is accessible to almost all sectors.

# Road to become a Data Scientist



# REFERENCES


- https://slideplayer.com/slide/10398517/
- https://www.slideshare.net/ryanorban/how-to-become-a-data-scientist
- Dhar, V. (2013). "Data science and prediction". Communications of the ACM. 56 (12): 64–73. doi:10.1145/2500499.
- Hayashi, Chikio (1 January 1998). "What is Data Science? Fundamental Concepts and a Heuristic Example". In Hayashi, Chikio; Yajima, Keiji; Bock, Hans-Hermann; Ohsumi, Noboru; Tanaka, Yutaka; Baba, Yasumasa (eds.). Data Science, Classification, and Related Methods. Studies in Classification, Data Analysis, and Knowledge Organization. Springer Japan. pp. 40–51. doi:10.1007/978-4-431-65950-1_3. ISBN 9784431702085.
- Davenport, Thomas H.; Patil, DJ (October 2012), Data Scientist: The Sexiest Job of the 21st Century, Harvard Business Review
- Jeff Leek (12 December 2013). "The key word in "Data Science" is not Data, it is Science". Simply Statistics.
- https://www.analyticsvidhya.com/blog/2015/09/applications-data-science/
- https://www.edureka.co/blog/data-science-applications/
- https://dutchdatascienceweek.nl/2018/04/05/the-impact-of-data-science-on-society/
- https://www.educba.com/data-science-and-its-growing-importance/